



- 1 The matrix $\mathbf{M}$ represents the sequence of two transformations in the x - y plane given by a stretch parallel to the x -axis, scale factor k ($k \neq 0$), followed by a shear, x -axis fixed, with $(0, 1)$ mapped to $(k, 1)$.

(a) Show that $\mathbf{M} = \begin{pmatrix} k & k \\ 0 & 1 \end{pmatrix}$. [4]

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- (b) The transformation represented by $\mathbf{M}$ has a line of invariant points.

Find, in terms of k , the equation of this line. [3]

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